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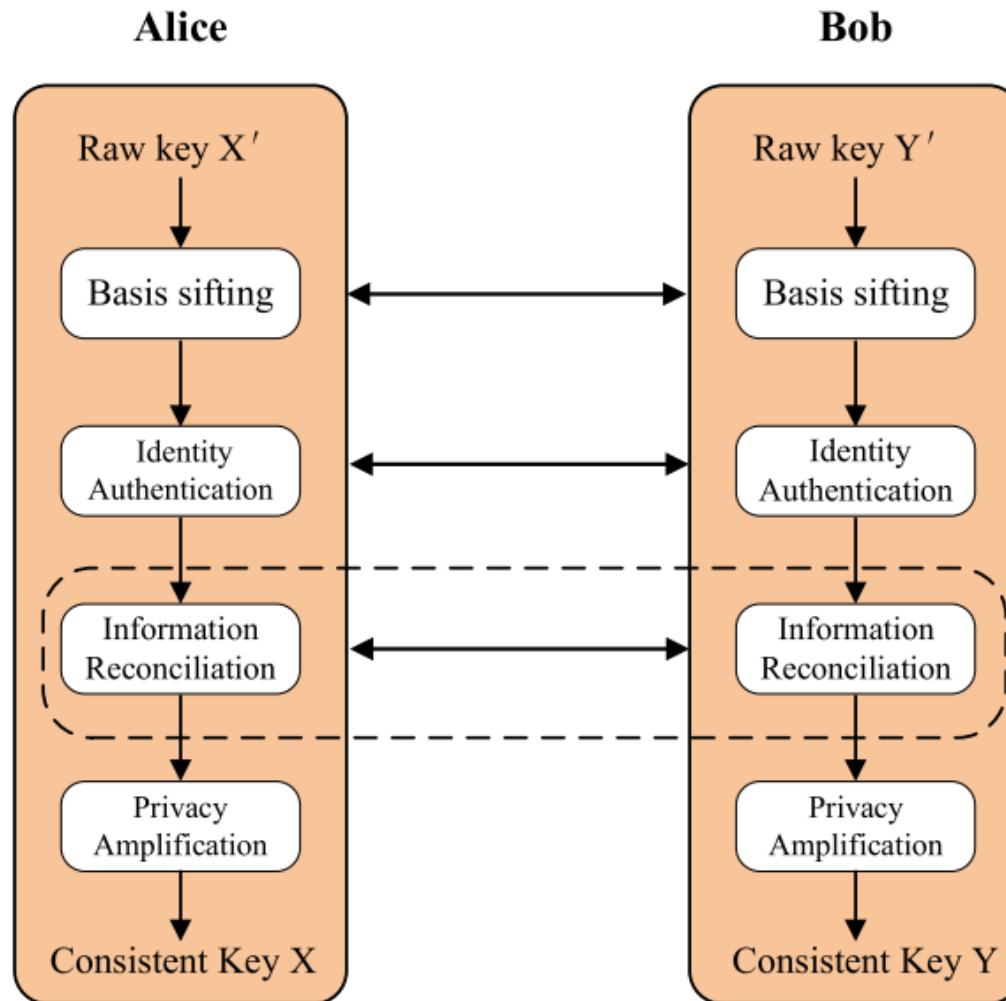
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High-Speed FPGA-Based Post-Processing Architecture for CV-QKD Systems

Information Reconciliation

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Information Reconciliation and Privacy Amplification



Target Platform: Xilinx Zynq-7000 System-on-Chip (SoC).

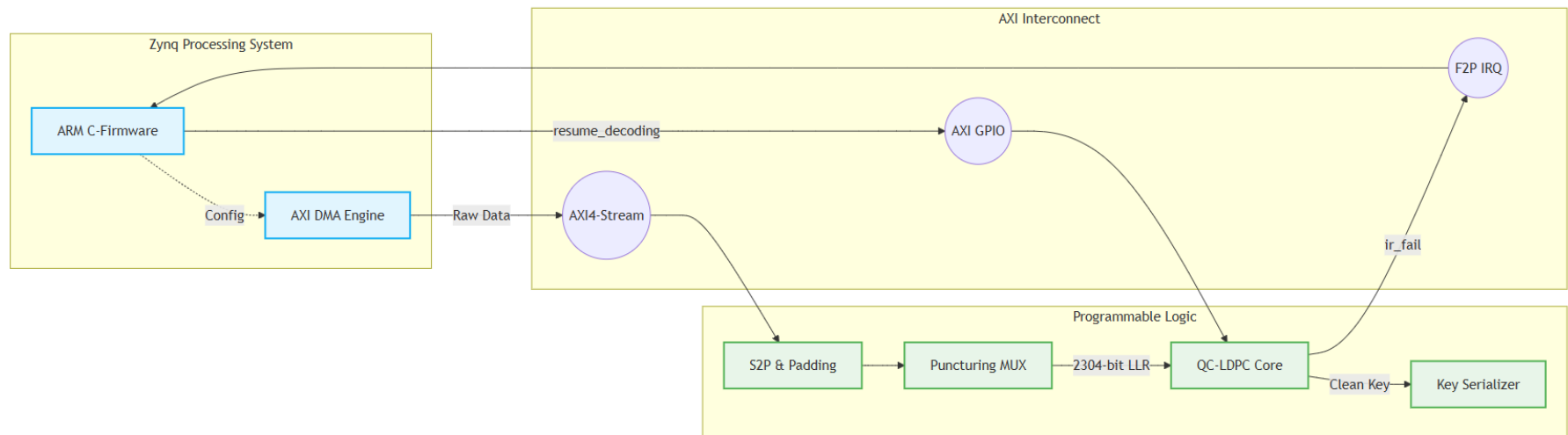
Architectural Approach: Hardware/Software (HW/SW) Co-Design.

- ✓ ARM Cortex-A9 (PS): Handles protocol orchestration, DMA data buffering, and dynamic syndrome padding.
- ✓ FPGA Fabric (PL): Accelerates computationally heavy mathematical operations.

Core Cryptographic Stages: Utilizes an IEEE 802.16e QC-LDPC Decoder

- Dynamic Blind Reconciliation FSM
- Early Termination Logic
- Offset Min-Sum (OMS) Calibration
- Hard Puncturing Multiplexing

HW/SW Co-Design Architecture



LDPC Core Technical Specifications

Parameter	Specification
Base Matrix Standard	IEEE 802.16e (WiMAX)
Block Length (N)	2304 bits
Expansion Factor (Zc)	96
Supported Code Rates	1/2, 2/3, 3/4 (Dynamic switching)
Decoding Algorithm	Offset Min-Sum (OMS)
Max Iterations	32 (With Early Termination capability)
Hardware Architecture	Partially Parallel (Zc nodes processed per clock cycle)

Fixed-Point Quantization Strategy

- Log-Likelihood Ratio (LLR) Input: 10-bit Signed
- Check-to-Variable (C2V) Messages: 8-bit Signed
- Variable-to-Check (V2C) Messages: 11-bit Signed

Zero-Overhead Rate Adaptation via Hard Puncturing

Challenge: Traditional Multi-Rate LDPC decoders require instantiating multiple distinct hardware cores or storing massive ROM lookup tables for different matrices, which rapidly depletes the FPGA's Look-Up Tables (LUTs) and Block RAMs (BRAMs).

Solution:

- Implemented a unified, single-core architecture based exclusively on the Rate 1/2 mother matrix.
- Developed a dynamic Hard Puncturing Multiplexer situated right before the decoder input.
- Depending on the PS control signal (`code_rate`), the MUX artificially gates specific blocks of incoming Log-Likelihood Ratios (LLRs) to 0.

Partially Parallel Architecture & Compare Tree

Challenge: A fully parallel instantiation of an $N=2304$ LDPC matrix would require an astronomical number of wiring interconnects, leading to severe routing congestion and inevitable timing closure failures on mid-range Zynq SoCs.

Innovative Solution:

- **Partially Parallel Processing:** Instead of computing all nodes simultaneously, the hardware processes exactly one sub-matrix block ($Z_c = 96$ nodes) per clock cycle.
- **Min-Tree Comparator (CNU):** The Check Node Units (CNU) in the Offset Min-Sum algorithm require finding the 1st and 2nd minimum values among heavily quantized incoming signals. By deploying a pipelined binary Compare-Tree, the logic depth was dramatically shortened.

Multi-Rate Blind Reconciliation FSM

The Quantum Channel Problem: The Quantum Bit Error Rate (QBER) in free-space or fiber optics fluctuates unpredictably. A rigid, fixed code rate leads to either inefficient key extraction (wasting syndrome) or catastrophic decoding failure.

The "Blind" Solution:

- **Aggressive Initial State:** The system attempts to decode the raw key using the highest possible Code Rate (e.g., Rate 3/4) to maximize the Secure Key Rate (SKR).
- **Hardware/Software Handshake:** If the LDPC core fails to converge, it instantly fires the `ir_fail_intr` hardware interrupt to the Zynq ARM processor.
- **Autonomous Down-Shifting:** The ARM processor injects additional Syndrome bits into the pipeline and fires a `resume_decoding` pulse.

Throughput Optimization via Early Termination

The Latency Problem: The Belief Propagation algorithm iteratively refines probabilities up to a worst-case boundary of $\text{MAX_ITER} = 32$. However, in high-quality optical links ($\text{QBER} < 2\%$), the matrix often resolves all errors within the first 5 to 8 iterations.

The Smart Halting Mechanism:

- Integrated a high-speed parity evaluation tree (`all_layers_parity_ok`) inside the Check Node Unit.
- Modified the core FSM in the `LAYER_WRITE` state to continuously monitor this flag.
- If the mathematical condition is completely satisfied before reaching `MAX_ITER`, the FSM short-circuits the iterative loop and jumps directly to the `OUTPUT_RES` state to flush the clean key out.

Bit-Accurate Golden Model Verification

The Belief Propagation algorithm iteratively Python Golden Model (hw_exact_sim.py): Developed a custom, bit-accurate Python simulator that perfectly mirrors the hardware's architecture. It replicates the exact Fixed-Point Quantization thresholds, Offset Min-Sum (OMS) approximations, and Hard Puncturing logic used in the FPGA.

Cross-Verification Strategy:

- The Verilog RTL simulation (tb_system_top.v) dumps intermediate layer computations and final decoded arrays.
- These dumps are aggressively cross-referenced against the Python Golden Model outputs.

Simulation Results & Verification

Block	Initial Errors	QBER	Remaining Errors	Fixed rate	State
0	89	3.86%	22	75.3%	Failed(Trapping Set)
1	76	3.30%	0	100%	Success
2	87	3.78%	0	100%	Success
3	79	3.43%	0	100%	Success
4	91	3.95%	0	100%	Success
5	194	8.42%	146	24.7%	Failed (Beyond Shannon Limit)

Simulation Results & Verification

- **Trapping Set Issue (Block 0):**

Even with fewer initial errors than Block 4, Block 0 fails to converge due to error distribution, not just QBER. The errors align with short cycles in the Tanner graph, forming a trapping set. This causes the Offset Min-Sum decoder (with 6-bit quantization) to get stuck in a local minimum, leaving residual errors.

- **Shannon Limit Issue (Block 5):**

Block 5 has a very high initial QBER (8.42%), exceeding the Shannon limit for the IEEE 802.16e LDPC (Rate 1/2). As a result, decoding failure is expected, since the noise level is beyond what the code can theoretically correct.

- Decoding failure is driven by both **error structure (trapping sets)** and **fundamental information-theoretic limits (Shannon bound)**—not just the raw number of errors.

Transmitter-Side Syndrome Generation (Altera DE2 Integration)

- Establish a robust communication link (e.g., Ethernet or UART/SPI) between the Altera DE2 (Alice) and the Xilinx Zynq (Bob) to transmit the 144-byte Syndrome data in real-time.
- This dual-FPGA setup will demonstrate a complete, physically separated QKD Information Reconciliation testbed.

Overcoming Trapping Sets & Error Floors

- Higher Quantization: Expand the LLR datapath from 6-bit to 8-bit fixed-point representation to minimize round-off losses, granting the decoder higher precision to escape local minimums during belief propagation.

A large, stylized graphic on the left side of the slide. It consists of a red background with a circular pattern of white dots of varying sizes, creating a sense of depth and movement. The word "HUST" is written in white, bold, sans-serif capital letters in the center of this graphic.

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THANK YOU !